MATH329 · OPERATIONS RESEARCH

Quiz 3-4 Practice

Big-M method · 30 minutes · term 471

Muharib

01 Questions

Answer the question in the space below it. The full solution is in a separate file.

Q STANDARD FORM + BIG-M

Given the following problem:

Maximize $Z = 2x_1 + 3x_2 - 5x_3$

Subject to $4x_1 + 3x_2 - x_3 \leq 24$

$x_1 + x_2 + 2x_3 \geq 18$

$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$

1. Write the problem in standard form "Equation form".

2. Solve the problem applying Big-M method.

Before you solve

Big-M did not appear in the past midterms. It comes only from a term 471 quiz.